

# Mnemosyne — Execution Plan A: The Memory System

Goal: the world's best-performing AI memory system

**Version:** 1.0 · **Date:** 2026-07-08 · **Status:** Proposed (awaiting go) **Scope:** the Mnemosyne engine and its capabilities ( [src/mnemosyne/](#) , plus engine-side [eval/](#) regression suites). **Companion doc:** *Execution Plan B — Benchmarking & the Leaderboard* owns measurement, publication, and the public leaderboard. This plan builds the capabilities; Plan B proves and publishes them. Where this plan says "measured/published," the authority is Plan B. **Audience:** an autonomous engineering agent (or fleet) executing end-to-end, plus human operators for gated evidence capture.

## 0. How to use this document (executing agent, read first)

Execution spec, not prose. Work top-to-bottom within a phase; respect the dependency map (§6). Every task has a stable ID ( [§1.2](#) ), a **Definition of Done (DoD)**, and required **artifacts**. Nothing is done without its artifacts.

**Hard guardrails (violating any = stop and escalate):**

- 1. **Do not break the §31 invariant rails or the 5 §33 test classes.** All existing [eval/](#) gates and [tests/](#) stay green. This work is **additive** to the attested v1.0 system.
- 2. **Capability first, claims later.** This plan makes the system *better*; it does not publish numbers. All external claims flow through Plan B's Public-Benchmark Publication Protocol (PBPP). No headline claims originate here.
- 3. **All perf/scale numbers require measured evidence** per the existing §11 "measurement-gap register." No asserted-but-unmeasured values.
- 4. **Provenance & capability checks apply to every new write path** (text, media, activation-level). Retrieved content is treated as data, never executed (§31 R6). New memory modalities inherit the same rails.
- 5. **Model-agnostic core.** Keep the minimal-dependency philosophy (core requires only [cryptography](#) ); new heavy deps go behind optional extras or the sidecar/ [services/](#) layer, never into core.

**Ground-truth references (source of truth over this doc if they conflict):** [docs/ARCHITECTURE-OVERVIEW.md](#) , [docs/ENGINE-CONTRACT.md](#) , [docs/blueprint/Mnemosyne-Performance-and-Refactoring-Blueprint.md](#) (Waves A–E, §9.2.7, §11), [.planning/STATE.md](#) , [.planning/codebase/{ARCHITECTURE,STACK,STRUCTURE}.md](#) , [eval/README.md](#) .

**DoD template:** *Code merged + all existing gates green + new regression cell added + artifacts written to the named path + one-paragraph result note in [eval/reports/](#) .*

## 1. Objective & success criteria (capability targets)

The system is "best-performing" when it hits these **capability** targets. Their *measurement and publication* are Plan B's job (via [eval/public/](#) + PBPP); this plan is responsible for making them true.

| #     | Capability                                  | Signal (measured by Plan B)                                    | Target                                                    |
|-------|---------------------------------------------|----------------------------------------------------------------|-----------------------------------------------------------|
| S-i   | Elite retrieval                             | Deterministic Recall@k / nDCG on public suites                 | Top-tier (≥ best published no-LLM score)                  |
| S-ii  | Multi-hop associative recall                | Deterministic R@2/@5 + EM/F1 on multi-hop sets                 | Meet/beat HippoRAG 2 baselines                            |
| S-iii | <b>Multi-hop answer synthesis (the gap)</b> | LLM-judged QA (disclosed reader)                               | Private <a href="#">qa_hard_v2</a> ≈ 0.62 → ≥ <b>0.85</b> |
| S-iv  | Security under attack                       | Attack-success-under-defense on MINJA/AgentPoison/ PoisonedRAG | Best-in-field; publishable                                |
| S-v   | Calibration                                 | ECE + abstention vs public labels                              | Best-in-field; publishable                                |

| #    | Capability        | Signal (measured by Plan B)                     | Target                 |
|------|-------------------|-------------------------------------------------|------------------------|
| S-vi | Credible at scale | Warm <b>and</b> concurrent P95; 100k-item cells | Measured distributions |

**What "best" does *not* mean here:** it does not mean winning a contested board like LoCoMo, and it does not mean any self-defined internal metric being presented as a public result. Those are Plan-B guardrails; this plan simply builds real capability.

## 2. The eclectic architecture synthesis (best ideas → mapped to what Mnemosyne already has)

Mnemosyne already implements a large share of the field's best ideas. This table is the design ledger; **priority = benchmark ROI**. Full source citations in Appendix A.

| Capability                   | Best external idea(s)                            | Source                             | Mnemosyne today                                                                               | Action                                                           | Prio       |
|------------------------------|--------------------------------------------------|------------------------------------|-----------------------------------------------------------------------------------------------|------------------------------------------------------------------|------------|
| Two-tier episodic ↔ semantic | Complementary Learning Systems                   | McClelland/Kumaran/Hassabis        | Evidence ledger (episodic) + typed projections (semantic) — already CLS-shaped                | <b>Keep</b> ; frame explicitly                                   | —          |
| Associative recall           | Graph index + Personalized PageRank              | HippoRAG / HippoRAG 2              | <b>Have it</b> ( <code>algorithms.ppr_power_iteration</code> , <code>graph_ppr_cache</code> ) | <b>Extend</b> (HippoRAG2 deeper passage integration)             | High       |
| Hierarchical sensemaking     | Community summaries                              | Microsoft GraphRAG                 | RAPTOR gist tree in <code>summarizer</code>                                                   | <b>Keep</b> ; add global map-reduce query mode                   | Med        |
| Multi-hop answer synthesis   | Iterative retrieve → read; reader model          | HippoRAG2, LongMemEval readers     | <b>Weak spot</b> ( <code>qa_hard_v2</code> ≈ 0.62 )                                           | <b>Build</b> — the #1 lever (S3, S1)                             | <b>Top</b> |
| Consolidation loop           | Offline replay + distillation ("sleep/dreaming") | Google "Sleep", OpenAI Dreaming    | 11-role warm loop ( <code>replayer...</code> <code>user_model_updater</code> )                | <b>Keep</b> ; add async batch "sleep" job                        | Med        |
| Multi-timescale memory       | Continuum Memory System (freq-staggered modules) | Google Nested Learning / Hope      | Single warm-loop cadence                                                                      | <b>Extend</b> — spectrum of update rates → anti-forgetting       | High       |
| Write gating                 | Surprise (gradient magnitude)                    | Titans; EM-LLM (Bayesian surprise) | Promotion gate + mutation budget                                                              | <b>Extend</b> — surprise signal into <code>promotion_gate</code> | Med        |
| Episodic segmentation        | Event boundaries + temporal-contiguity retrieval | EM-LLM                             | Bitemporal, but no event-segmented recall                                                     | <b>Add</b> — episode boundaries + temporal-adjacency reads       | High       |
| Forgetting / deletion        | One-shot selective forget; adaptive decay        | Larimar; Titans; ACT-R             | Fidelity tiers + crypto-shred + optional ACT-R decay                                          | <b>Keep</b> (already best-in-class)                              | —          |
| Durable parametric facts     | Localized weight edits; sparse KV memory layers  | MEMIT; Meta Memory Layers          | Parametric tier (LoRA/TTT, <code>parametric.py</code> )                                       | <b>Extend</b> — evaluate MEMIT-style edits as a write path       | Low        |
| Compiled context             | Offline-trained reusable KV cache                | Cartridges (Stanford)              | None                                                                                          | <b>Research</b> — per-tenant "cartridge" for hot corpora         | Med        |

| Capability                                     | Best external idea(s)                                          | Source                                      | Mnemosyne today                       | Action                                                        | Prio |
|------------------------------------------------|----------------------------------------------------------------|---------------------------------------------|---------------------------------------|---------------------------------------------------------------|------|
| Retrieval/param split                          | Small model + huge swappable DB                                | RETRO                                       | Already retrieval-first over Postgres | <b>Keep</b>                                                   | —    |
| Bounded working memory                         | Compressive per-layer memory; memory tokens                    | Infini-attention; RMT                       | N/A (model-agnostic)                  | <b>Skip</b> (model-internal, out of scope)                    | —    |
| <b>Activation-space memory + introspection</b> | J-space / global workspace, persona vectors, SAE features      | <b>Anthropic</b> (§3.5)                     | None                                  | <b>Research track</b> — calibration + security differentiator | Med  |
| Agent memory API contract                      | <code>/memories</code> file ops + context editing + compaction | Anthropic memory tool / context engineering | <code>mnome</code> CLI + 48 MCP tools | <b>Align</b> the MCP memory surface to this contract          | Med  |

### 3. Phased build

#### S1 — Close the multi-hop answer-synthesis gap (top priority, critical path)

Retrieval is already strong; synthesis (~0.62) is the ceiling on every QA benchmark. This is the single highest-ROI capability.

- **S1.1 Iterative retrieve → read loop.** Query decomposition → PPR spreading-activation hops over the existing graph → evidence assembly, reusing the hybrid retriever (dense + BM25 + PPR + RRF/MMR). Borrow HippoRAG 2's deeper passage integration.
- **S1.2 Grounded reader/synthesis step.** A disclosed self-hosted role-LLM (Ollama) composes an answer *only* from retrieved, provenance-tagged evidence; fail-closed, retrieved text handled as data (§31 R6). Every claim in the answer must trace to an evidence CID.
- **S1.3 Episode-aware recall.** Add EM-LLM-style event boundaries + temporal-contiguity reads so synthesis pulls coherent episodes, not fragments.
- **DoD:** `qa_hard_v2` and (via Plan B) public LongMemEval-QA ≥ 0.85; per-hop retrieval traces prove groundedness; no regression on deterministic recall. Targets S-iii.

#### S2 — Capability upgrades (by benchmark ROI)

- **S2.1 Multi-timescale consolidation spectrum** (Nested Learning CMS): parameterize the warm loop into fast/medium/slow cadences so new writes never overwrite semantic structure at once. DoD: continual-learning regression shows reduced forgetting.
- **S2.2 Async "sleep" consolidation job** (Google Sleep / OpenAI Dreaming): offline replay + distillation of durable facts into slower tiers, plus **temporal freshness / auto-expiry** so facts age correctly ("going to X" → "went to X"). DoD: staleness metric + freshness regression test.
- **S2.3 Global map-reduce query mode** (GraphRAG) over the RAPTOR tree for "themes across everything" questions. DoD: sensemaking eval cell.
- **S2.4 Surprise-gated writes** (Titans/EM-LLM): route a surprise signal into `promotion_gate` so high-information events are preferentially consolidated. DoD: write-precision/recall cell.
- **S2.5 (Research) Cartridges** — offline-compiled per-tenant KV cache for hot corpora. DoD: latency/throughput A/B vs retrieval baseline; go/no-go note.

#### S3 — Security & calibration capabilities (differentiators)

Build the capability here; **publish via Plan B** (these fill columns no competitor fills). - **S3.1 Expand the adversarial corpus.** Grow the 59-attack `poison_suite.json` to full MINJA + AgentPoison + PoisonedRAG threat models; implement/verify

defenses so the metric is **attack-success-under-defense**, not just block-rate. DoD: adversarial report + regression gate. - **S3.2 Calibration hardening.** Ensure the conformal-calibration/abstain layer produces reliability diagrams + ECE that hold up against *public* benchmark labels (not only the private set). DoD: reliability artifacts consumable by Plan B's bundle.

**S4 — Performance / scale close-out (Phase 9 / Waves B–D)**

Clear the §11 measurement-gap register so capability claims are credible at scale: - **S4.1** Provider bake-off + default selection (TEI vs sidecar vs ONNX) — resolve `eval/provider_bakeoff/` (protocol exists, results don't). - **S4.2** Null-embedding production backfill evidence. - **S4.3** halfvec / pgvectorscale DiskANN; 100k-item benchmark cells. - **S4.4 Concurrent** P95 (not just warm-serial 149.5 ms) — resolve the known CPU-embed bottleneck. - **DoD:** measured distributions in `eval/latency*` and `eval/provider_bakeoff/` ; §11 register closed. Targets S-vi.

**S5 — Research track: activation-space memory & introspection (off critical path)**

See §3.5 (below). Requires white-box access to open-weight models — feasible via the self-hosted Ollama role-LLMs. Explicitly labeled research; not gating Goal-A capability targets.

**3.5 The Anthropic introspection / "J-space" track (novel differentiator)**

Anthropic's 2025–26 interpretability line is directly usable — with strict **epistemic humility**: we make *functional* claims, never consciousness claims.

- **J-lens / global workspace** (Anthropic, Jul 2026; `github.com/anthropics/jacobian-lens` ): a readable list of "words on the model's mind but not said." **Use:** log a compact J-lens readout as an activation-trace on the consolidation path; use it as a **poisoning/deception tripwire** (evaluation-awareness, fabrication markers).
- **Introspective awareness** (Lindsey, Oct 2025) via concept injection: self-reports are a **weak (~20% reliable), calibratable** signal. **Use:** feed gated introspective confidence into the conformal-calibration/abstain layer — never as sole ground truth.
- **Persona vectors** (Anthropic, Aug 2025; arXiv 2507.21509): activation directions for traits via contrastive prompting. **Use:** rolling projection = drift-detection for the user-model/consolidation loop; screen candidate memories/adaptor data before persistence.
- **SAE features** (Scaling Monosemanticity): stable inspectable concept vocabulary. **Use:** tag memories by active interpretable features → activation-space retrieval + audit.
- **Dual-use warning:** concept injection *is* memory poisoning (the false-intention "bread" experiment). Any activation-level write path carries provenance + capability checks exactly like text (guardrail #4).
- **DoD:** `research/activation-memory/` spike with (1) J-lens tripwire on the Ollama role-LLM, (2) a persona-vector drift metric in the ops dashboard, (3) a written go/no-go on promoting any of it to core.

**4. Cross-cutting: verification & quality**

- Every capability change adds a **regression cell** to `eval/` and keeps all §33 classes green.
- No capability is "done" on a single run — point estimates carry Wilson/bootstrap CIs (existing harness convention).
- Capability numbers stay **internal** until Plan B's PBPP + third-party reproduction clear them for any external claim.

**5. Risk register (system-specific)**

| Risk                                   | Mitigation                                                                                                 |
|----------------------------------------|------------------------------------------------------------------------------------------------------------|
| Multi-hop synthesis underperforms 0.85 | S1 is critical path + gated; if short, ship deterministic-retrieval strength and iterate — never overclaim |
| New modality breaks a §31 rail         | Additive-only; every write path re-runs the rail regression before merge                                   |
| Heavy deps creep into core             | Sidecar/optional-extra discipline (guardrail #5)                                                           |

| Risk                                                 | Mitigation                                                          |
|------------------------------------------------------|---------------------------------------------------------------------|
| Activation track over-interpreted as "consciousness" | Functional-signal framing only; research-labeled; off critical path |
| Scale numbers asserted not measured                  | §11 register must be closed (S4) before any scale claim             |

## 6. Timeline, roles, dependency map

**Critical path (bold):** 1. **S1 multi-hop synthesis** — weeks 1–6 (top lever). 2. S2 capability upgrades || S3 security/calibration — weeks 3–9. 3. **S4 perf/scale close-out** — weeks 5–11. 4. S5 / §3.5 activation research — parallel, off critical path.

**Dependencies:** S1 unblocks S-iii; S3/S4 feed Plan B's artifact bundle; the public harness that *measures* all of this is **Plan B §3 (eval/public/)** — coordinate the shared eval core (serialize writes to it).

**Roles:** *core-eng agent* (S1–S2), *ml agent* (S1 reader, S5 research), *sec agent* (S3.1), *infra agent* (S4). Run independent phases as parallel sub-agents.

### Definition-of-Done checklist (Plan A)

- [ ] Multi-hop QA ≥ 0.85 with grounded per-hop traces (S1).
- [ ] Multi-timescale consolidation + async sleep job + freshness/expiry live (S2).
- [ ] Global map-reduce query mode + surprise-gated writes shipped (S2).
- [ ] Adversarial corpus expanded; attack-success-under-defense capability ready for publication (S3.1).
- [ ] Calibration reliability artifacts hold vs public labels (S3.2).
- [ ] §11 measurement-gap register closed: concurrent+warm P95, 100k cells, provider default (S4).
- [ ] Activation-memory research spike + go/no-go note (S5).
- [ ] All §31 rails and §33 classes green throughout.

## Appendix A — Frontier-lab source map (architecture ideas)

Titans (arXiv 2501.00663), Infini-attention (2404.07143), Nested Learning/Hope (2512.24695), Google "Sleep" (2606.03979), Memorizing Transformers (2203.08913), RETRO (2112.04426), Meta Memory Layers (2412.09764), Larimar/IBM (2403.11901), Microsoft GraphRAG (2404.16130), MEMIT (2210.07229), HippoRAG / HippoRAG 2 (2405.14831 / 2502.14802), EM-LLM (2407.09450), Test-Time Training (2407.04620), Cartridges (2506.06266), CLS theory (Kumaran/Hassabis/McClelland 2016). **Anthropic:** global workspace / J-lens (transformer-circuits.pub/2026/workspace; github.com/anthropics/jacobian-lens), introspective awareness (transformer-circuits.pub/2025/introspection), persona vectors (2507.21509), scaling monosemanticity (2024), memory tool + context engineering (platform.claude.com docs; anthropic.com/engineering/effective-context-engineering-for-ai-agents).

## Appendix B — Dependency on Plan B

Plan B (*Benchmarking & the Leaderboard*) owns: the `eval/public/` harness that measures S-i...S-vi, the Public-Benchmark Publication Protocol (PBPP) that governs any external claim about this system, the reproducibility artifact bundle, third-party reproduction, and the neutral leaderboard on which Mnemosyne is entered. This plan produces capabilities; Plan B proves and publishes them.

*This plan is additive to the attested v1.0 system. It does not modify the §31 rails or §33 gates.*